



Faculty of Computers and Artificial Intelligence

Cairo University

IS211: Introduction to Database System

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جامعة القاهرة

Project ID: 103 Title: Course Management System Program: General Section Group: S23, 24

Course Management System

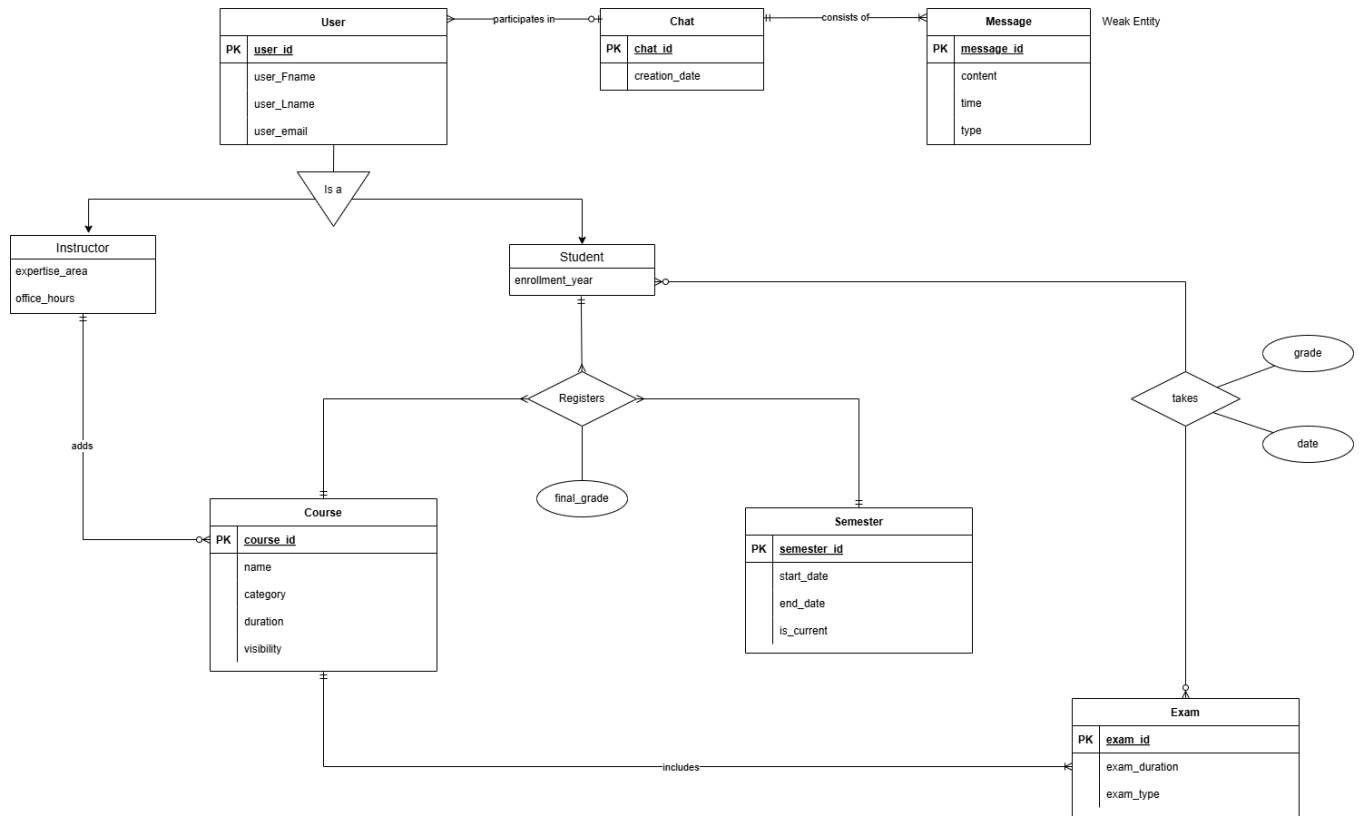
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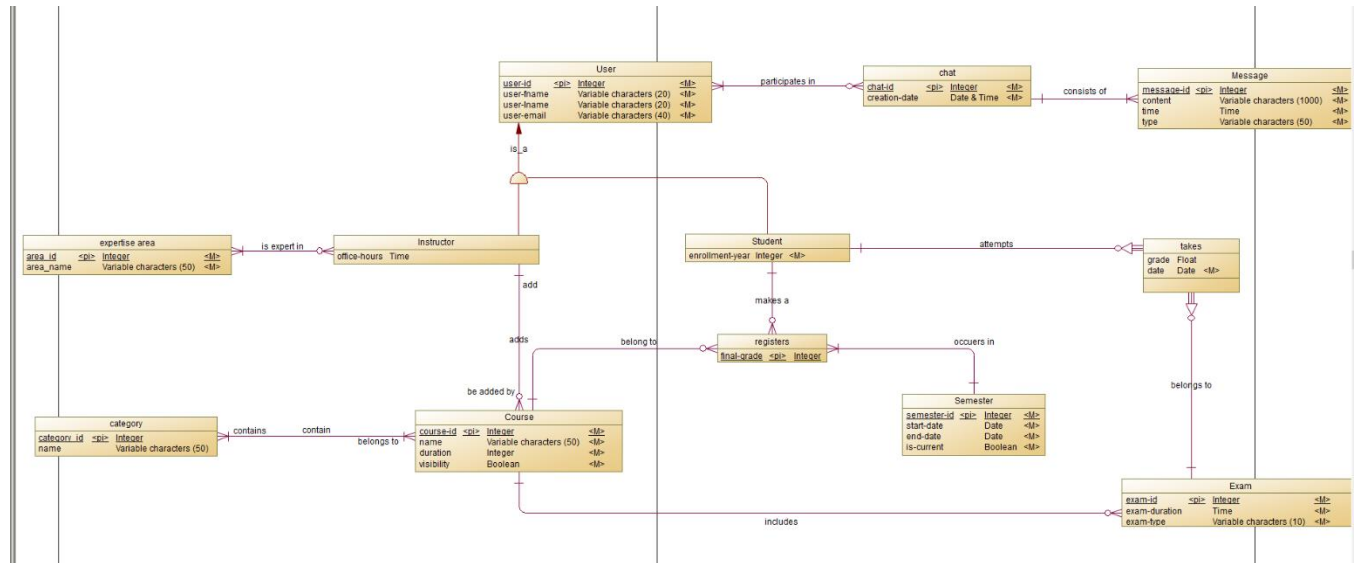
Conceptual ERD



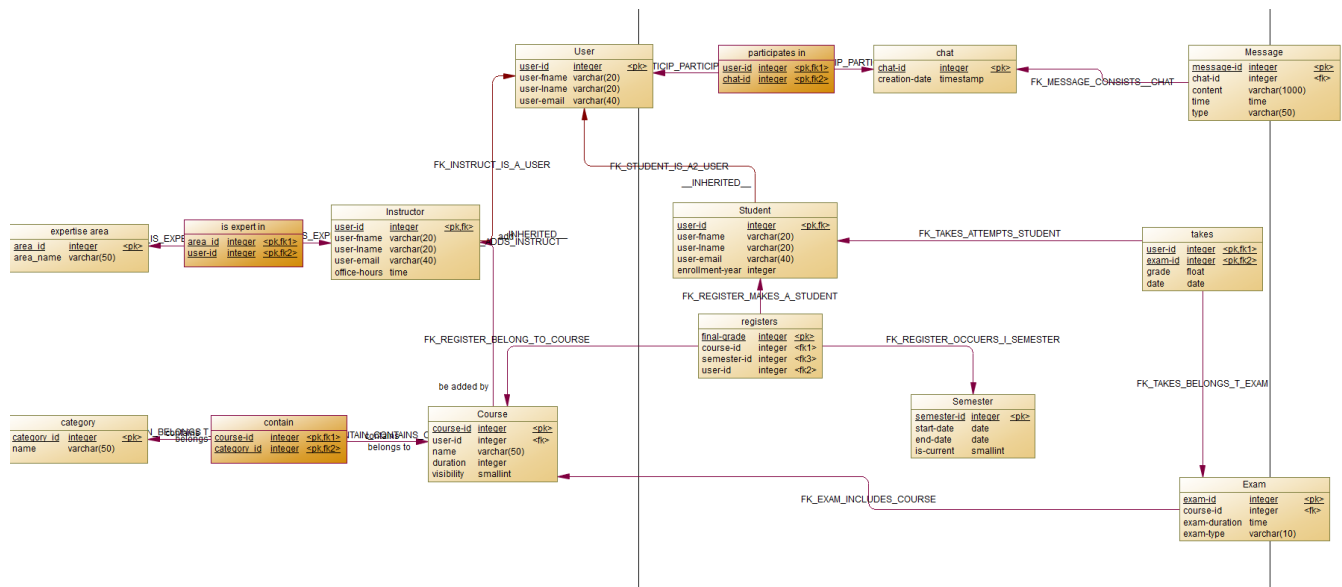
For a full visualization of this ERD, click here:

https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000FF&edit=_blank&layers=1&nav=1&title=ERD.drawio&dark=auto#Uhttps%3A%2F%2Fdrive.google.com%2Fuc%3Fid%3D1Jt0V1TOg4eT9noC8LiqeAZN0KbCRtGD%26export%3Ddownload

Conceptual Model in Power Designer



Physical ERD



Requirements

These requirements describe what the system should support, based on the entities and relationships in our conceptual model.

1. User Functionalities

- The system must allow signing up a new user as either a **student** or an **instructor**.
- A user must have a unique user_id and include personal details such as first name, last name, and email.
- A user can only play **one role** (either student or instructor), as shown in the specialization relation

2. Course Management

- An **instructor** can create and manage multiple **courses**.
- Each course must include:
 - A unique course ID
 - Course name, one or multiple categories, duration, and visibility status (e.g., hidden or visible to students).
 - Visibility status is set by instructor
- Courses can be **offered in multiple semesters** (many-to-many relationship)

3. Semester Management

- The system maintains records of **semesters**, each with:
 - A unique semester_id
 - Start and end dates
 - A flag (is_current) to indicate whether the semester is active (current semester).

4. Course Registration

- A **student** can register in multiple **courses**, and a **course** can have many **registered students**.
- Each registration must be recorded **per semester** and include:
 - user_id (student)
 - course_id
 - semester_id
 - Final course grade (if available)

5. Exam Management

- An **exam** must belong to one **course**.
- Each course may have **multiple exams** (e.g., midterm, final, quizzes, exercises).
- Exam attributes include:
 - exam_date
 - exam_duration
 - exam_type (e.g., midterm, final, quizzes, exercises)

6. Student Exam Participation

- Students can take multiple exams.
- Each **exam participation record** must track:
 - Which **student** took which **exam**
 - The **grade** the student received on that exam

7. Chatting

- Each student can have a live chat request to instructors
- Instructors see the request and answer student questions
- Each chat have must have one or many messages